

BOOKPILE - Personal Library Manager

Project Goal

Personal web application to manage a physical home library.

The main objective is not only to know which books I own, but also to know exactly where each book is physically located.

The application will be used by a single user.

Technology preference:

- FastAPI
- SQLite
- React + Vite + TypeScript

The project should be designed incrementally, starting with a minimal but extensible architecture.

Version 1 Scope

Books

Each book contains:

- id
- title
- author
- status
- goodreads_url (optional)
- notes (optional)

Book status:

- PENDING
 - READ
-

Dashboard

Display:

- Total books
 - Pending books
 - Read books
-

CRUD Operations

Required operations:

- Create book
 - Edit book
 - Delete book
 - Change status
-

Physical Library Model

The application must support the physical organization of books.

Hierarchy:

Bookcase → Shelf → Container → Book

Bookcase

Represents a physical bookcase.

Examples:

- Office Bookcase
- Living Room Bookcase

Fields:

- id
 - name
 - description (optional)
-

Shelf

Represents a shelf inside a bookcase.

Shelves are numbered from top to bottom.

Example:

Shelf 1 = top shelf

Fields:

- id

- bookcase_id
 - shelf_number
-

Container

Represents a group of books within a shelf.

Container types:

- ROW
- PILE

Container layers:

- BACKGROUND
- FOREGROUND

Containers are numbered:

- left to right
- top to bottom

Fields:

- id
- shelf_id
- container_type
- layer
- container_number

Examples:

- Background vertical row
 - Background horizontal pile
 - Foreground horizontal pile
-

Book Location

Every book should be locatable.

Fields:

- book_id
- container_id
- position

Position numbering:

- left to right

- top to bottom

Examples:

Bookcase: Office Shelf: 3 Container: Background Row Position: 7

or

Bookcase: Office Shelf: 4 Container: Foreground Pile 2 Position: 3

Future Features (ordered by priority)

1. Reading suggestions
 2. Random pending book
 3. Oldest pending book
 4. Book acquisition tracking
 5. acquisition_date
 6. read_date
 7. time spent in pending pile
 8. Visual shelf representation
 9. graphical display of bookcases, shelves and containers
 10. Goodreads integration
 11. links
 12. ratings
 13. reviews
 14. Cover images
 15. OCR / camera-based book recognition
 16. very low priority
 17. only after the core application is fully functional
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Design Principles

- Simple UI
- Fast data entry
- Physical location is a first-class concept
- Single-user application
- Local-first architecture
- Extensible data model
- Avoid overengineering